

高考终于结束了，送给大家一道线性代数题暖暖场

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Abstract. 随着 2026 年高考生物结束铃的打响，我结束了三年的高中生活。终于高考结束了，这个暑假有时间回来研究图形学了。大的文章还在施工，作为回归礼，先送给各位一道线性代数题。本题的背景是下篇文章要解析的论文。

Keywords: 线性代数, Math, Computer Graphics, 数学题

1 Problem

Let $S^2 = \{\omega \in \mathbb{R}^3 : \|\omega\| = 1\}$ be the unit sphere equipped with its standard solid angle measure $d\omega$. Let $D_o : S^2 \rightarrow \mathbb{R}$ be a density function on the sphere, and let $M \in \mathbb{R}^{3 \times 3}$ be an invertible matrix.

Define a mapping $T : S^2 \rightarrow S^2$ by

$$T(\omega) = \omega_o = \frac{M^{-1}\omega}{\|M^{-1}\omega\|} \quad (1)$$

Suppose a transformed spherical density $D : S^2 \rightarrow \mathbb{R}$ is defined by preserving density with respect to solid angle, namely

$$D(\omega)d\omega = D_o(\omega_o)d\omega \quad (2)$$

Prove that

$$D(\omega) = D_o\left(\frac{M^{-1}\omega}{\|M^{-1}\omega\|}\right) \frac{|\det(M^{-1})|}{\|M^{-1}\omega\|^3} \quad (3)$$

Here $d\omega$ and $d\omega_o$ denotes infinitesimal solid angle elements on S^2 .

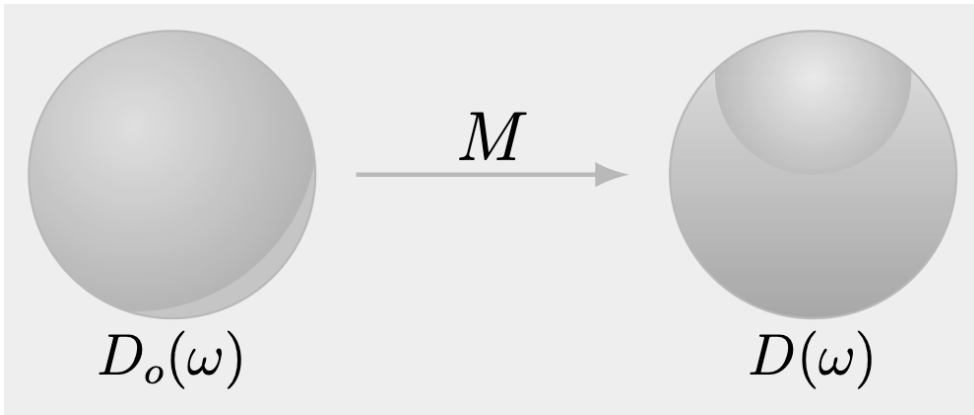


图 1. The transform of the matrix M

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